Least squares problem: $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$

$y = f(x)$ ← unknown.

$\{f_1, f_2, \dots, f_k\}$, $\quad f \cong \alpha_1 f_1 + \alpha_2 f_2 + \cdots + \alpha_k f_k$

$$\min_{\alpha \in \mathbb{R}^k} \|\Phi\alpha - y\|_2^2 \quad (1)$$

$\hat{\alpha}$ is a minimizer iff

$$\Phi^T\Phi\,\hat{\alpha} = \Phi^T y$$: Normal system of eqns

To solve (1) is equivalent to solving the system of eqns

$$\Phi^T\Phi\,\alpha = \Phi^T y$$

$k \times k$ system

$\Phi^T\Phi$ is invertible (Φ is a full column rank matrix)

i) Row operations on $\Phi^T\Phi$ (Gaussian elimination)

ii) LU decomposition of a square/invertible.

Given $A \in \mathbb{R}^{k \times k}$ invertible.

Want to find $Q \in \mathbb{R}^{k \times k}$ orthogonal ($Q^T Q = I = Q Q^T$)

and an upper triangular matrix $R \in \mathbb{R}^{k \times k}$ s.t.

$$A = QR$$

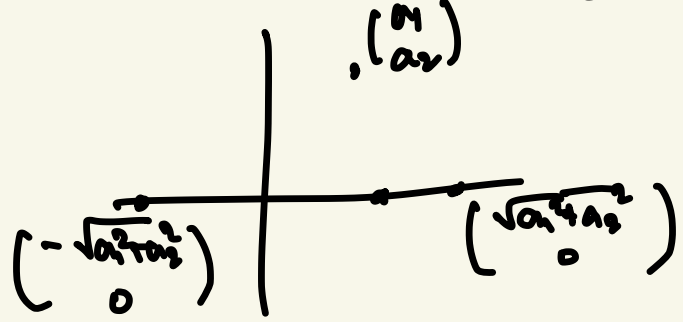
Solving: $Ax = y$, $QRx = y \Rightarrow \underbrace{Rx}_{\text{upper triangular}} = \boxed{Q^T y}$

(back substitution)

Rotators:

2×2 case

$\mathbb{R}^2$



$$\|Qx\|_2^2 = \|x\|_2^2$$

$$(Qx)^T Qx \qquad x^T x$$

$$x^T (Q^T Q) x$$

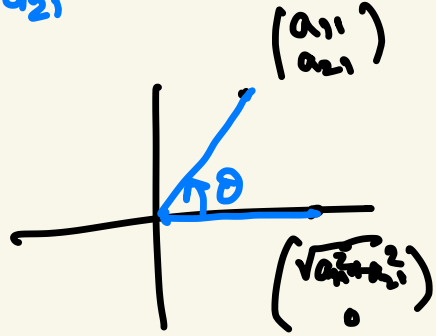
$A \in \mathbb{R}^{2\times 2}$

$$Q\,A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \longrightarrow \begin{bmatrix} * & * \\ 0 & * \end{bmatrix}$$

R

$$(*) = \sqrt{a_{11}^2 + a_{21}^2}$$

$$\frac{1}{\sqrt{a_{11}^2+a_{21}^2}} \begin{bmatrix} a_{11} & a_{21} \\ -a_{21} & a_{11} \end{bmatrix} \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} \sqrt{a_{11}^2+a_{21}^2} \\ 0 \end{bmatrix}$$



orthogonal

$a_{11} = r\cos\theta$

$a_{21} = r\sin\theta$

$r = \sqrt{a_{11}^2+a_{21}^2}$

$$Q^T \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} = \begin{pmatrix} \sqrt{a_{11}^2+a_{21}^2} \\ 0 \end{pmatrix} \Rightarrow Q^T A = \begin{bmatrix} \sqrt{a_{11}^2+a_{21}^2} & * \\ 0 & * \end{bmatrix} // R$$

$Ax = b$ $\quad A \in \mathbb{R}^{2\times 2}$ invertible.

$b \in \mathbb{R}^2$

$Q^T A x = Q^T b$

$\underbrace{\quad}$

$Rx = Q^T b$

Back-substitution

→ Generalize to 3×3, $A \in \mathbb{R}^{3\times 3}$ invertible.

$Ax = b$

$$\begin{bmatrix} * & 0 & * \\ 0 & 1 & 0 \\ * & 0 & * \end{bmatrix} \begin{bmatrix} q_{11} & q_{12} & 0 \\ q_{21} & q_{22} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} * & * & * \\ 0 & * & * \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Q_2^T $\quad Q_1^T$ $\quad A$

$$\underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{Q}\underbrace{\begin{bmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_k \\ | & | & & | \end{bmatrix}}_{A} = \begin{bmatrix} | & | & & | \\ Qa_1 & Qa_2 & \cdots & Qa_k \\ | & | & & | \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} - & q_1^T & - \\ - & q_2^T & - \\ & \vdots & \\ - & q_k^T & - \end{bmatrix}}_{Q}\underbrace{\begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}}_{A} = \begin{bmatrix} - & q_1^T A & - \\ - & q_2^T A & - \\ & \vdots & \\ - & q_k^T A & - \end{bmatrix}$$

$$\begin{bmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_k \\ | & | & & | \end{bmatrix}\begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix} = x_1 a_1 + x_2 a_2 + \cdots + x_k a_k$$

$$\begin{bmatrix} q_{11} & \cdots & q_{1k} \\ q_{21} & \cdots & q_{2k} \\ \vdots & & \\ q_{n1} & \cdots & q_{nk} \end{bmatrix} \begin{bmatrix} a_{11} & \cdots & a_{1k} \\ a_{21} & \cdots & a_{2k} \\ \vdots & & \\ a_{k1} & \cdots & a_{kk} \end{bmatrix}$$

$$= \begin{bmatrix} q_{11}a_{11} + q_{12}a_{21} + \cdots q_{1k}a_{k1} & - & - \\ q_{21}a_{11} + q_{22}a_{21} + \cdots + q_{2k}a_{k1} & - & \cdots & - \\ \vdots & & & \\ q_{k1}a_{11} + q_{k2}a_{21} + \cdots + q_{kk}a_{k1} & - & & - \end{bmatrix}$$

$$= \begin{bmatrix} Qa_1 \end{bmatrix}$$

$$\begin{bmatrix} \sqrt{a_{11}^2+a_{21}^2} & 0 & a_{31} \\ \frac{}{r} & & \frac{}{r} \\ 0 & 1 & 0 \\ -a_{31} & 0 & \sqrt{a_{11}^2+a_{21}^2} \\ \frac{}{r} & & \frac{}{r} \end{bmatrix} \begin{bmatrix} a_{11}/\sqrt{} & a_{21}/\sqrt{} & 0 \\ -a_{21}/\sqrt{} & a_{11}/\sqrt{} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$r = \sqrt{a_{11}^2+a_{21}^2+a_{31}^2}$$

$$= \begin{bmatrix} \sqrt{a_{11}^2+a_{21}^2} & * & * \\ 0 & * & * \\ a_{31} & * & * \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{a_{11}^2+a_{21}^2+a_{31}^2} & * & * \\ 0 & * & * \\ 0 & * & * \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & \end{bmatrix} Q_2^T Q_1^T A = \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & * & * \end{bmatrix} = \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{bmatrix}$$

$$Q_{32}^T \; Q_{31}^T \; Q_{21}^T A = \begin{bmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{bmatrix}$$

R

→ Solving the system

$$\underbrace{Q_{32}^T Q_{31}^T Q_{21}^T A}_{} x = Q_{32}^T Q_{31}^T Q_{21}^T b$$

$$\underbrace{Rx}_{} = \underbrace{Q_{32}^T Q_{31}^T Q_{21}^T b}_{} \in \mathbb{R}^3$$

$$Q_{32}^T Q_{31}^T Q_{21}^T A = R$$

$$\Rightarrow A = \underbrace{Q_{21} Q_{31} Q_{32}} R$$

$$\Rightarrow A = QR \quad \leftarrow \text{QR decomposition of } A.$$

$$\begin{bmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_k \\ | & | & & | \end{bmatrix}_{k \times k} = \underbrace{\begin{bmatrix} | & | & & | \\ q_1 & q_2 & \cdots & q_k \\ | & | & & | \end{bmatrix}}_{Q} \underbrace{\begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1k} \\ 0 & r_{22} & \cdots & r_{2k} \\ \vdots & & & \\ 0 & 0 & \cdots & r_{kk} \end{bmatrix}}_{R}$$

A invertible.

$$A = \begin{bmatrix} | & | & & | \\ a_1 & a_2 & \cdots & a_k \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ q_1 & q_2 & \cdots & q_k \\ | & | & & | \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1k} \\ 0 & r_{22} & \cdots & r_{2k} \\ \vdots & \vdots & & \vdots \\ 0 & 0 & & r_{kk} \end{bmatrix}$$

Q R

r_1 : 1st column of R

r_2 : 2nd column of R

r_k : k^{th} column of R

$$a_1 = Q r_1$$

$$a_1 = r_{11} q_1$$

$$a_2 = Q r_2 = r_{12} q_1 + r_{22} q_2$$

$$a_3 = Q r_3 = r_{13} q_1 + r_{23} q_2 + r_{33} q_3$$

$$\vdots$$

$$a_k = Q r_k = r_{1k} q_1 + r_{2k} q_2 + \cdots + r_{kk} q_k$$

Gram - Schmidt orthogonalization

$\Rightarrow A = [a_1 \cdots a_n]$

$\{a_1, a_2, \dots, a_k\}$ linearly independent vectors.

Find orthonormal vectors $q_1, q_2, \dots, q_k$ s.t.

$$\text{span}\{a_1\} = \text{span}\{q_1\}$$

$$\text{span}\{a_1, a_2\} = \text{span}\{q_1, q_2\}$$

$$\vdots$$

$$\text{span}\{a_1, a_2, \dots, a_k\} = \text{span}\{q_1, q_2, \dots, q_k\}$$

$$q_1 = \frac{a_1}{\|a_1\|} \qquad r_1 = \|a_1\| \quad \Rightarrow \quad a_1 = r_{11} q_1$$

$$q_2 = \frac{a_2 - \langle a_2, q_1 \rangle q_1}{\|a_2 - \langle a_2, q_1 \rangle q_1\|} \quad \Rightarrow \quad a_2 = \|a_2 - \langle a_2, q_1 \rangle q_1\| q_2 + \langle a_2, q_1 \rangle q_1$$

$$\begin{bmatrix} 3 & 1 \\ 0 & 1 \end{bmatrix}$$

a_1 a_2

$a_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$, $a_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

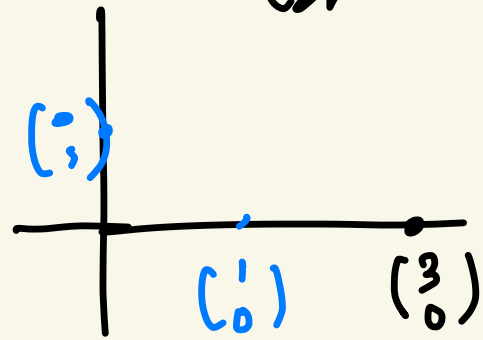
q_1, q_2

$\cdot \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$q_1^T q_1 = 1$

$q_2^T q_2 = 1$

$q_1^T q_2 = 0$



$a_1 = \begin{pmatrix} 3 \\ 0 \end{pmatrix}$

$q_1 = \frac{a_1}{\|a_1\|} = \frac{1}{3}\begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$q_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$

$\frac{\begin{pmatrix} 0 \\ 2 \end{pmatrix}}{\left\| \begin{pmatrix} 0 \\ 2 \end{pmatrix} \right\|} = \frac{1}{2}\begin{pmatrix} 0 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$